

Resolución parcial curso K3521

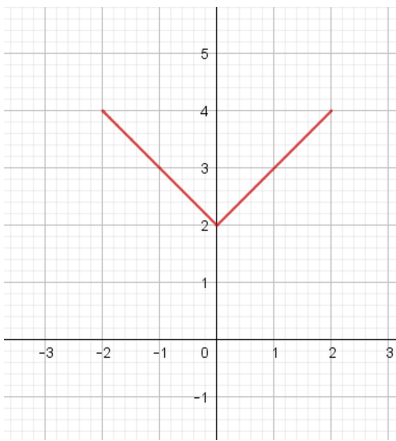
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Ejercicio 1

Dada la función $f(t) = \begin{cases} t+2 & t \in (0,2) \\ \dots & t \in (\dots, \dots) \end{cases} f(t) = f(t+4)$

1. Completar la función en todo el eje real para que su STF sea solo de cosenos y desarrollarla.
2. Usando el desarrollo del ítem anterior hallar los C_n de la SEF
3. Con el resultado obtenido en el punto anterior hallar el valor al cual converge $\sum_{k=0}^{\infty} \frac{1}{(2k+1)^2}$

$$1) f(t) = \begin{cases} t+2 & t \in (0,2) \\ -t+2 & t \in (-2,0) \end{cases} f(t) = f(t+4) \quad b_n = 0 \quad T=4 \\ L=2 \quad \omega_0 = \frac{\pi}{2}$$



$$\frac{a_0}{2} = 3 \quad \text{Se ve en el gráfico}$$

$$a_n = \frac{2}{L} \int_0^L f(t) \cos(n \omega_0 t) dt$$

$$a_n = \frac{2}{2} \int_0^2 (t+2) \cos(n \frac{\pi}{2} t) dt$$

$$a_n = \int_0^2 t \cos(n \frac{\pi}{2} t) dt + 2 \int_0^2 \cos(n \frac{\pi}{2} t) dt$$

$$a_n = \left[\frac{\cos(n \frac{\pi}{2} t)}{n^2 \frac{\pi^2}{4}} + \frac{t \sin(n \frac{\pi}{2} t)}{n \frac{\pi}{2}} + \frac{2 \sin(n \frac{\pi}{2} t)}{n \frac{\pi}{2}} \right]_0^2 = 4 \left(\frac{(-1)^n - 1}{n^2 \pi^2} \right)$$

$$a_n = 0 \quad n \text{ par}$$

$$a_n = \frac{-8}{n^2 \pi^2} \quad n \text{ impar}$$

$$S(t) = 3 - \frac{8}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \cos((2k+1) \frac{\pi}{2} t)$$

$$2) C_n = \frac{a_n}{2} \quad \text{por ser } f(t) \text{ par} \Rightarrow C_n = \frac{-4}{n^2 \pi^2} \quad n \text{ impar} \\ C_n = 0 \quad n \text{ par}$$

$$3) S(t) = 3 - \frac{8}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} \cos((2k+1) \frac{\pi}{2} t)$$

$$S(0) = f(0) = 2 \Rightarrow 3 - \frac{8}{\pi^2} \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = 2 \Rightarrow \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = (3-2) \cdot \frac{\pi^2}{8}$$

$$\therefore \sum_{k=0}^{\infty} \frac{1}{(2k+1)^2} = \frac{\pi^2}{8}$$

Ejercicio 2

Resuelva la siguiente ecuación integral usando Transformada de Laplace:

$$y(t) - \int_0^t y(u) \cdot \sin(t-u) du = t^4$$

$$\mathcal{L}(y(t)) - \mathcal{L}\left(\int_0^t y(u) \sin(t-u) du\right) = \mathcal{L}(t^4)$$

$$Y(s) - Y(s) \frac{1}{s^2+1} = \frac{24}{s^5}$$

$$Y(s) \left(\frac{s^2+1}{s^2+1} - \frac{1}{s^2+1} \right) = \frac{24}{s^5} \Rightarrow Y(s) = \frac{24(s^2+1)}{s^7} = \frac{24}{s^5} + \frac{24}{s^7}$$

$$y(t) = \mathcal{L}^{-1}\left(\frac{24}{s^5}\right) + \frac{\mathcal{L}^{-1}\left(\frac{24}{s^7}\right)}{30} = t^4 + \frac{t^6}{30}$$

$$y(t) = t^4 + \frac{t^6}{30}$$

Ejercicio 3

Dado los siguientes conjuntos:

$$A = \{z \in \mathbb{C} : z^3 + (2-2j)z^2 + z(3+2j) = 0\}$$

$$B = \{\text{Valor Principal de } \ln(z) : z = j^{1-3j}\}$$

$$C = \{z \in \mathbb{C} : \left| \frac{z+3-2j}{1+3j} \right| > 1\}$$

Hallar los elementos de A, B, C y $(A \cup B) \cap C$

$$\sqrt{-3-4j} = \begin{cases} x = \sqrt{\frac{5-3}{2}} = \pm 1 \\ y = \sqrt{\frac{3+3}{2}} = \pm 2 \end{cases}$$

$$A = \{0; -j; -2+3j\}$$

$$2. B \rightarrow \ln(z)/z = j^{1-3j}$$

$$1) A \rightarrow z(z^2 + (2-2j)z + (3+2j)) = 0$$

$$\therefore z=0 \rightarrow z_1=0 \text{ es raíz}$$

Resuelvo la ecuación cuadrática

$$z_{2,3} = \frac{-(2-2j) \pm \sqrt{(2-2j)^2 - 4(3+2j)}}{2}$$

$$z_{2,3} = \frac{-2+2j \pm \sqrt{4-8j-4-12-8j}}{2}$$

$$z_{2,3} = \frac{-2+2j \pm \sqrt{-12-16j}}{2} = \frac{-2+2j \pm 2\sqrt{-3-4j}}{2}$$

$$z_2 = -1+j-2j = -j$$

$$z_3 = -1+j-1+2j = -2+3j$$

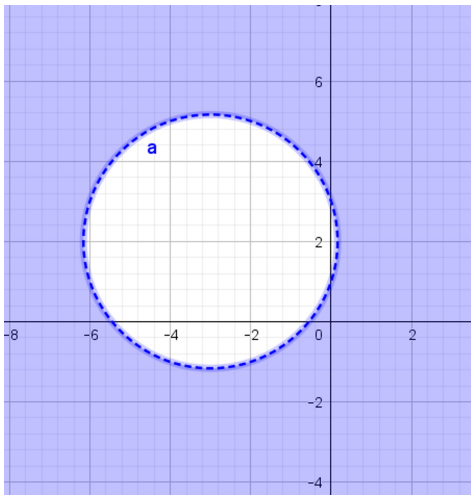
$$\rightarrow \ln(z) = (1-3j)\ln(j) = (1-3j)\frac{\pi}{2}j$$

$$\ln(z) = \frac{\pi}{2}j + 3\frac{\pi}{2} \therefore B = \left\{ 3\frac{\pi}{2} + \frac{\pi}{2}j \right\}$$

$$C \rightarrow \left| \frac{z+3-2j}{1+3j} \right| > 1 \rightarrow \frac{|z+3-2j|}{|1+3j|} > 1 \rightarrow \frac{\sqrt{(x+3)^2 + (y-2)^2}}{\sqrt{10}} > 1$$

$$\therefore C = \{(x+3)^2 + (y-2)^2 > 10\}$$

$$\therefore C = \{(x+3)^2 + (y-2)^2 > 10\}$$



$$(A \cup B) \cap C = \left\{ 0, -j, \frac{3\sqrt{2}}{2} + \frac{\pi}{2}j \right\}$$

Ejercicio 4

Resolver la siguiente ecuación en diferencia usando transformada Z:

$$2x(n+2) - x(n+1) = 3x(n) - 1, x(0) = 0, x(1) = 1$$

Usando transformada Z

$$2 \cdot Z(x(n+2)) - Z(x(n+1)) = 3Z(x(n)) - Z(1)$$

$$2(Z^2 X(z) - \cancel{Z^2 x(0)} - \cancel{Z x(1)}) - (Z X(z) - \cancel{Z x(0)}) = 3X(z) - \frac{Z}{z-1}$$

$$2Z^2 X(z) - 2Z - Z X(z) - 3X(z) = -\frac{Z}{z-1}$$

$$X(z) \underbrace{(2Z^2 - Z - 3)}_{(z+1)(2z-3)} = -\frac{Z}{z-1} + 2Z$$

$$(z+1)(2z-3)$$

$$\sigma$$

$$2(z+1)(z-3/2)$$

$$X(z) = \frac{-Z + 2Z(z-1)}{(z-1)(z+1)(2z-3)} = z \left(\frac{-1 + 2z - 2}{(z-1)(z+1)(2z-3)} \right) = z \left(\frac{\cancel{2z-3}}{(z-1)(z+1)\cancel{(2z-3)}} \right)$$

$$X(z) = z \left(\frac{A \leftarrow \frac{1}{2}}{z-1} + \frac{B \leftarrow \frac{1}{2}}{z+1} \right) \rightarrow X(z) = \frac{1}{2} \cdot \frac{z}{z-1} - \frac{1}{2} \frac{z}{z+1}$$

$$x(n) = \frac{1}{2} - \frac{1}{2}(-1)^n$$

Ejercicio 5

Dada la función de transferencia $G(s) = \frac{17(s^2 + s - k)}{(s-2)(s^2 + 8s + 17)}$

- Hallar el valor de k para que $G(s)$ sea estable
Sabido que la entrada $f(t)$ es un escalón unitario $E(t)$:
- Dibuje un corte por el eje real de $Y(s)$
- Halle la respuesta temporal $y(t)$.

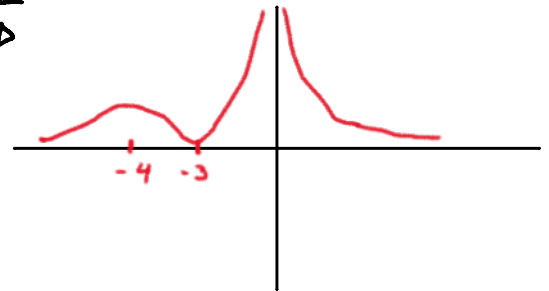
a) 2 polos positivos $\therefore$ Hay que simplificar
 $k=6 \therefore 2^2 + 2 - k < 0 \therefore k=6$

ENTRADA $E(t) \rightarrow F(s) = \frac{1}{s}$ ya que $E(t) = 1$

$$b) Y(s) = G(s) \cdot F(s) \Rightarrow Y(s) = \frac{17(s^2 + s - 6)}{(s-2)(s^2 + 8s + 17)} \cdot \frac{1}{s}$$

$$Y(s) = \frac{17(\cancel{s-2})(s+3)}{(\cancel{s-2})(s^2 + 8s + 17)} \cdot \frac{1}{s}$$

$$Y(s) = \frac{17(s+3)}{s \cdot (s^2 + 8s + 17)}$$



$$c) y(t) = ?$$

$$Y(s) = \frac{17(s+3)}{s \cdot (s^2 + 8s + 17)} = \frac{A}{s} + \frac{B \cdot s + C}{s^2 + 8s + 17}$$

$\frac{17 \cdot 3}{17} = 3$
 $\swarrow \quad \searrow \quad \nwarrow$
 $A \quad B \quad C$

$$\therefore 3(s^2 + 8s + 17) + (Bs + C)s = 17s + 51$$

$$3 + B = 0 \rightarrow B = -3$$

$$24 + C = 17 \rightarrow C = -7$$

$$Y(s) = \frac{3}{s} + \frac{-3s - 7}{s^2 + 8s + 17} = \frac{3}{s} - 3 \frac{(s+4-4)}{(s+4)^2 + 1} - \frac{7}{(s+4)^2 + 1}$$

$$Y(s) = \frac{3}{s} - \frac{3(s+4)}{(s+4)^2 + 1} + \frac{5}{(s+4)^2 + 1} \therefore$$

$$y(t) = 3 - 3 \cos(t) e^{-4t} + 5 \sin(t) e^{-4t}$$